

Unimodal Function and Steepest Descent Method

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1. Unimodal Function

Monotonic Increasing and Decreasing Functions

Definition: A real-valued function $f(x)$ defined on an interval I is said to be:

- **Monotonic increasing** if for all $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) \leq f(x_2).$$

- **Strictly increasing** if for all $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) < f(x_2).$$

- **Monotonic decreasing** if for all $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) \geq f(x_2).$$

- **Strictly decreasing** if for all $x_1, x_2 \in I$,

$$x_1 < x_2 \implies f(x_1) > f(x_2).$$

Examples:

1. $f(x) = 3x + 2$ is strictly increasing on $\mathbb{R}$.
2. $f(x) = -2x + 1$ is strictly decreasing on $\mathbb{R}$.
3. $f(x) = c$ (a constant) is both non-increasing and non-decreasing.

Definition: A function $f(x)$ defined on an interval $[a, b]$ is said to be **unimodal** if it has only one local minimum (or maximum) in that interval. That is, there exists a point x^* such that

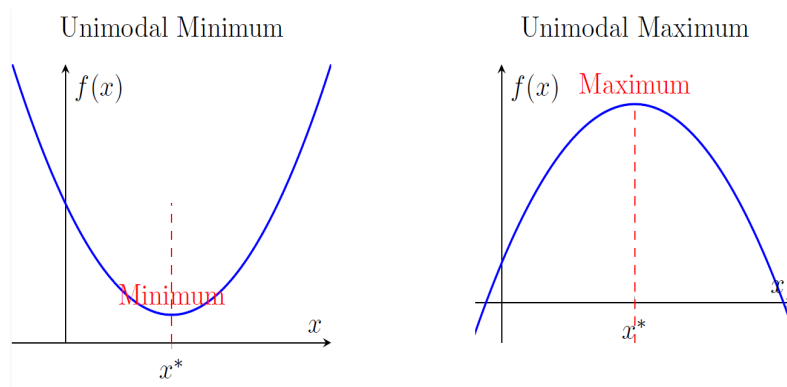
$$\begin{cases} f(x) \text{ is strictly decreasing for } x < x^*, \\ f(x) \text{ is strictly increasing for } x > x^*. \end{cases}$$

Hence, x^* is the unique minimum point of $f(x)$ on $[a, b]$.

Example: The quadratic function $f(x) = (x-2)^2 + 1$ is unimodal on $\mathbb{R}$ with a unique minimum at $x^* = 2$.

Graphical Interpretation: A unimodal function has a single “valley” (for a minimization problem) or a single “peak” (for maximization).

Remark: Unimodality is an important assumption in *line search methods* such as the *steepest descent method* and *golden section search*.



Necessary and Sufficient Conditions for an Extremal Point

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable function.

Necessary Condition (First-Order Condition)

If x^* is a local extremum (minimum or maximum) of f , then

$$\nabla f(x^*) = 0.$$

That is, all first partial derivatives of f vanish at x^* .

Sufficient Condition (Second-Order Condition)

Suppose f is twice differentiable, and $\nabla f(x^*) = 0$. Let the *Hessian matrix* of f at x^* be

$$H(x^*) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

Then:

- If $H(x^*)$ is **positive definite**, x^* is a **strict local minimum**.
- If $H(x^*)$ is **negative definite**, x^* is a **strict local maximum**.
- If $H(x^*)$ is **indefinite**, x^* is a **saddle point**.

Remark 1 (Test for Definiteness of a Matrix) Let H be a symmetric matrix, and let

$$\Delta_1 = h_{11}, \quad \Delta_2 = \begin{vmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{vmatrix}, \quad \dots, \quad \Delta_n = \det(H)$$

be its leading principal minors. Then:

$$\begin{cases} H \text{ is } \textbf{positive definite} & \text{if } \Delta_k > 0 \text{ for all } k = 1, 2, \dots, n, \\ H \text{ is } \textbf{negative definite} & \text{if } (-1)^k \Delta_k > 0 \text{ for all } k = 1, 2, \dots, n, \\ H \text{ is } \textbf{indefinite} & \text{if } \Delta_k \text{ changes sign.} \end{cases}$$

2. Steepest Descent Method

Objective: To minimize a differentiable function $f(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$.

$$\text{Minimize } f(\mathbf{x}) \quad \text{where } f : \mathbb{R}^n \rightarrow \mathbb{R}$$

2.1 Principle

The **steepest descent method** (also called the *gradient descent method*) is an iterative optimization algorithm that moves the current solution $\mathbf{x}_k$ in the direction of the *negative gradient*, which points toward the direction of the maximum rate of decrease of f .

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \lambda_k \nabla f(\mathbf{x}_k)$$

where:

- $\nabla f(\mathbf{x}_k)$ is the gradient (vector of partial derivatives),
- $\lambda_k > 0$ is the step length or learning rate, usually determined by a line search.

2.2 Algorithmic Steps

Step 1: Choose an initial point $\mathbf{x}_0$ and a convergence tolerance $\varepsilon > 0$.

Step 2: Compute the gradient $\nabla f(\mathbf{x}_k)$.

Step 4: Determine the step length λ_k that minimizes

$$\phi(\lambda) = f(\mathbf{x}_k - \lambda \nabla f(\mathbf{x}_k))$$

over $\alpha > 0$.

Step 5: Update the point:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \lambda_k \nabla f(\mathbf{x}_k)$$

Step 3: If $\|\nabla f(\mathbf{x}_k)\| < \varepsilon$ or $|f(\mathbf{x}_{k+1}) - f(\mathbf{x}_k)| < \epsilon$, stop; $\mathbf{x}_{k+1}$ is the approximate optimal point.

Step 6: Set $k = k + 1$ and repeat from Step 2.

2.3 Convergence

The method converges to a local minimum if $f(\mathbf{x})$ is continuously differentiable and convex. For quadratic functions, convergence is guaranteed and often geometric.

2.4 Example

Problem 1 Use steepest descent method to find the

$$\text{Minimize } f(x_1, x_2) = 2x_1^2 + 2x_1x_2 + x_2^2 + x_1 - x_2.$$

such that the error can not be exceed by 0.5 with initial approximation $(0, 0)$.

Answer 1 Let us denote the initial approximation $X_0 = (x_1^0, x_2^0) = (0, 0)$. Then

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{bmatrix}, \implies [\nabla f]_{X_0} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

So, the first iteration is given by:

$$X_1 = X_0 - \lambda [\nabla f]_{X_0} = \begin{bmatrix} -\lambda \\ \lambda \end{bmatrix}$$

Therefore,

$$f(X_1) = 2(-\lambda)^2 + 2(-\lambda)\lambda + \lambda^2 + (-\lambda) - \lambda = \lambda^2 - 2\lambda := \phi(\lambda)$$

The value of λ can be calculated from

$$\phi'(\lambda) = 0 \implies \lambda = 1.$$

Hence, the approximate solution after the first iteration

$$X_1 = (x_1^1, x_2^1) = (-1, 1).$$

Since,

$$|f(X_1) - f(X_0)| = |(-1) - 0| = 1 \not\leq 0.5.$$

so we need to perform the further iteration.

$$[\nabla f]_{X_1} = \begin{bmatrix} -4 + 2 + 1 \\ -2 + 2 - 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

So, the first iteration is given by:

$$X_2 = X_1 - \lambda [\nabla f]_{X_1} = \begin{bmatrix} \lambda - 1 \\ \lambda + 1 \end{bmatrix}$$

Therefore,

$$f(X_2) = 2(\lambda - 1)^2 + 2(\lambda - 1)(\lambda + 1) + (\lambda + 1)^2 + (\lambda - 1) - (\lambda + 1) = 5\lambda^2 - 2\lambda - 1 := \psi(\lambda)$$

The value of λ can be calculated from

$$\psi'(\lambda) = 0 \implies \lambda = 1/5.$$

Hence, the approximate solution after the second iteration

$$X_2 = (x_1^2, x_2^2) = (-4/5, 6/5).$$

Since,

$$|f(X_2) - f(X_0)| = \left| -\frac{6}{5} - (-1) \right| = 0.2 < 0.5.$$

Hence, we achieved the optimal iteration. The optimal solution is

$$x_1 = -4/5, x_2 = 6/5, f_{opt} = -6/5.$$

Exercises

Problem 2 Use steepest descent method to find the

$$\text{Minimize } f(x_1, x_2) = x_1^2 - x_1x_2 + x_2^2.$$

such that the error can not be exceed by 0.5 with initial approximation $(1, 1/2)$.

Answer 2 $x_1 = 1/4, x_2 = 1/8, f_{opt} = 3/64$.